

MA2108S

A. Countability

- If $S = \bigcup_{n=1}^{\infty} S_n$, S_n finite, then S is countable.
- If $S_1, \dots, S_n$ are countable, then $S = \bigcup_{j=1}^n S_j$ is countable.
- If $S_1, S_2, \dots$ are countable, then $S = \bigcup_{j=1}^{\infty} S_j$ is countable.
- Schröder-Bernstein**: If there exist injections $f: A \rightarrow B$ and $g: B \rightarrow A$, then there exists bijection $h: A \rightarrow B$.
- Tut**: Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of disjoint finite open intervals. If $\mathcal{A}$ is infinite, then $\mathcal{A}$ is countable.
- Tut**: Set of algebraic numbers is countable.

B. Completeness of Real Numbers

B.1. Supremum property

- $\beta = \sup E$ if (1) β is an upper bound of E , (2) for any $\varepsilon > 0$, $\beta - \varepsilon$ is not upper bound of E .
- $\beta = \inf E$ if (1) β is an lower bound of E , (2) for any $\varepsilon > 0$, $\beta + \varepsilon$ is not lower bound of E .
- An ordered set S has **supremum property** if every subset bounded above has a supremum (in S).

B.2. Archimedean property

- Archimedean property**: $\forall x, y > 0, \exists n \in \mathbb{N}$ s.t. $nx > y$.
- Floor**: $\forall x \in \mathbb{R}, \exists n \in \mathbb{Z}$ s.t. $n \leq x < n+1$. Denote $n = \lfloor x \rfloor$
- For any $x > 0$, there exists $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < x$.
- $\mathbb{Q}$ **dense** in $\mathbb{R}$: $\forall a < b, \exists r \in \mathbb{Q}$ s.t. $a < r < b$.

B.3. Nested intervals

- Nested interval property**: $I_n = [a_n, b_n]$ nested, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.
- If $|I_n| \rightarrow 0$, then $\bigcap_{n=1}^{\infty} I_n = \{\xi\}$ is a singleton.
- $\mathbb{R}$ is uncountable.
Proof: Contradiction. Construct nested intervals s.t. each avoids one more point of $\mathbb{R}$ than the previous. Then apply nested interval property.

C. Sequences and Series

C.1. Limits of sequences

- Let $\{a_n\}$ be a sequence. Then $a_n \rightarrow A$ if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad |a_n - A| < \varepsilon.$$
- Upper bound**: Convergent $\Rightarrow$ bounded
- Lower bound**: If $a_n \rightarrow A$, $a_n \neq 0, A \neq 0$, then $\exists C > 0$ s.t. $|a_n| \geq C$.
- Order preserving**: If $a_n \rightarrow A, b_n \rightarrow B$, $a_n \geq b_n$, then $A \geq B$.
- Limit laws**: Hold if $\{a_n\}, \{b_n\}$ conv
- Evaluate limit of rational function: divide by the highest power
- Evaluate limit of square root: multiply conjugate
- Evaluate limit of power function: write in the form $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{f(n)}\right)^{f(n)}$, bound $m \leq f(n) < m+1$, squeeze.
- Squeeze**: If $a_n \leq b_n \leq c_n$ and $a_n, c_n \rightarrow A$, then $b_n \rightarrow A$.
- $a^{1/n} \rightarrow 1, n^{1/n} \rightarrow 1$
Stirling's approximation: $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$
- Stolz thrm**: Let $\{a_n\}$ be sequence, $\{b_n\}$ strictly increasing and div. If $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = A$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = A$.
Tip: Use if numerator/denominator is a summation
- Cauchy's first limit thrm**: If $a_n \rightarrow A$, then $\lim_{n \rightarrow \infty} \frac{a_1 + \dots + a_n}{n} = A$

C.2. Monotone convergence theorem

- MCT**: If $\{a_n\}$ increasing [decreasing], bounded above [below], then conv.
- For recursively defined sequences, use induction to show (1) increasing/decreasing; (2) bounded above/below. Then take $n \rightarrow \infty$ to find the limit.
- Euler's number e** : $e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.
- $a_n = \left(1 + \frac{1}{n}\right)^n \nearrow e \quad b_n = \left(1 + \frac{1}{n}\right)^{n+1} \searrow e$

C.3. Subsequences, Bolzano-Weierstrass theorem

- $\{b_k = a_{n_k} : k \in \mathbb{N}\}$ is a **subsequence** of $\{a_n\}$
- If $\{a_n\}$ conv, then all subsequences conv to the same limit.
- If $\{a_n\}$ has two subsequences with different limits, then $\{a_n\}$ div.
- Monotone subsequence thrm**: Any sequence has monotone subsequence.
Proof: a_n is a peak if $a_n \geq a_m$ for all $m \geq n$. Consider cases: infinitely / finitely many peaks, construct decreasing / increasing subsequence.
- Bolzano-Weierstrass**: Any bounded sequence has conv subsequence
Proof 1: Monotone subsequence
Proof 2: Nested interval
Tip: Useful to get convergence for bounded sequence

C.4. Cauchy sequences

- $\{a_n\}$ is a **Cauchy sequence** if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad |a_m - a_n| < \varepsilon.$$
- Convergent $\iff$ Cauchy
- $\{a_n\}$ is **contractive** if $\exists C \in (0, 1)$ s.t. $|a_{n+2} - a_{n+1}| \leq C|a_{n+1} - a_n|$
- Contractive $\implies$ Cauchy $\implies$ convergent
- Alternating harmonic series** $\sum \frac{(-1)^{n-1}}{n}$ conv

C.5. Properly divergent sequences

- $\lim_{n \rightarrow \infty} a_n = +\infty$ if
$$\forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n > C.$$
- $\lim_{n \rightarrow \infty} a_n = -\infty$ if
$$\forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n < C.$$
- $\lim_{n \rightarrow \infty} a_n = \infty$ if $\{|a_n|\}$ tends to $+\infty$.
- Squeeze**: If $a_n \geq b_n$ and $b_n \rightarrow +\infty$, then $a_n \rightarrow +\infty$
If $a_n \geq b_n$ and $a_n \rightarrow -\infty$, then $b_n \rightarrow -\infty$
- Reciprocals**: If $a_n \neq 0$ and $a_n \rightarrow \infty$, then $\frac{1}{a_n} \rightarrow 0$
If $a_n \neq 0$ and $a_n \rightarrow 0$, then $\frac{1}{a_n} \rightarrow \infty$

C.6. Upper and lower limits

- Upper limit** $\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup\{a_k : k \geq n\}$
Lower limit $\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \inf\{a_k : k \geq n\}$
- A is a **limit point** of $\{a_n\}$ if A is the limit of some subsequence
- Let $\{a_n\}$ be bounded; let E denote the set of limit points of $\{a_n\}$. Then both upper limit and lower limit of $\{a_n\}$ are contained in E . Moreover,
$$\limsup a_n = \max E, \quad \liminf a_n = \min E.$$
- For any bounded sequence $\{a_n\}$, $\liminf a_n \leq \limsup a_n$, where equality holds if and only if $\{a_n\}$ conv.
- Order preserving**: If $\{a_n\}, \{b_n\}$ bounded, $a_n \leq b_n$, then
$$\liminf a_n \leq \liminf b_n, \quad \limsup a_n \leq \limsup b_n.$$
- Tut**: If $\{a_n\}$ bounded, $b_n = \frac{a_1 + \dots + a_n}{n}$, then $\limsup b_n \leq \limsup a_n$.

C.7. Series and convergence tests

- Given sequence $\{a_n\}$, define sequence of partial sums $s_n = \sum_{k=1}^n a_k$. If $s_n \rightarrow A$, then $\sum a_n = A$.
- Divergence test**: If $\sum a_n$ convergent, then $a_n \rightarrow 0$.
Contrapositive: if $a_n \not\rightarrow 0$, then $\sum a_n$ divergent.
- Cauchy criterion**: $\sum a_n$ converges iff Cauchy:
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m > n \geq N, \quad \left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

Tip: Useful if limit is not known.
- MCT**: If $a_n \geq 0$, partial sums $\sum a_n$ bounded above, then $\sum a_n$ conv.
- $\sum a_n$ **abs conv** if $\sum |a_n|$ conv
- $\sum a_n$ **cond conv** if conv but not abs conv
- Abs conv $\implies$ conv
- Comparison test**: If $0 \leq a_n \leq \lambda b_n$, then $\sum b_n$ conv $\implies \sum a_n$ conv
- For $0 < p \leq 1$, $\frac{1}{n^p} \geq \frac{1}{n} > 0$, so $\sum \frac{1}{n^p}$ div.
- For $p \geq 2$, $0 < \frac{1}{n^p} \leq \frac{1}{n^2}$, so $\sum \frac{1}{n^p}$ conv.
- Limit comparison test**: Let $\lambda = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.
 - If $\lambda > 0$, then $\sum a_n$ conv $\iff \sum b_n$ conv.
 - If $\lambda = 0$, then $\sum b_n$ conv $\implies \sum a_n$ conv; $\sum a_n$ div $\implies \sum b_n$ div.
- p-series**: For $p > 1$, $\sum \frac{1}{n^p}$ conv.

- Proof*: Dyadic partition, group terms of series as $\sum_{n=1}^{2^m} a_n = \sum_{k=1}^m \sum_{n=2^{k-1}}^{2^k-1} a_n$
E.g. **harmonic series** $\sum \frac{1}{n}$ div, **2-series** $\sum \frac{1}{n^2}$ conv
- Cauchy condensation**: $a_n > 0$ decr, then $\sum a_n$ conv $\iff \sum 2^n a_{2^n}$ conv
Proof: Dyadic partition
Tip: Useful to deal with denominator $\log_2 n$
- Leibniz test**: If $a_n \geq 0, a_n \searrow 0$, then $\sum (-1)^{n+1} a_n$ conv
E.g. $\sum \frac{(-1)^n}{\sqrt{n}}$ conv
- Root test**:
 - If $|a_n|^{1/n} \leq r$ for all $n \geq n_0$, where $0 \leq r < 1$, then $\sum a_n$ abs conv
 - If $|a_n|^{1/n} \geq 1$ for all $n \geq n_0$, then $\sum a_n$ div

- Suppose $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = r \geq 0$.
 - If $r < 1$, then $\sum a_n$ abs conv
 - If $r > 1$, then $\sum a_n$ div
- Ratio test**: Suppose $a_n \neq 0$
 - If $\left|\frac{a_{n+1}}{a_n}\right| \leq r$ for all $n \geq n_0$, where $0 < r < 1$, then $\sum a_n$ abs conv
 - If $\left|\frac{a_{n+1}}{a_n}\right| \geq 1$ for all $n \geq n_0$, then $\sum a_n$ div
- Suppose $a_n \neq 0$ and $\lim_{n \rightarrow \infty} \left|\frac{a_{n+1}}{a_n}\right| = r \geq 0$.
 - If $r < 1$, then $\sum a_n$ is absolutely convergent.
 - If $r > 1$, then $\sum a_n$ is divergent.
 - If $r = 1$, then the test is inconclusive.
- $\sum_{n=0}^{\infty} \frac{1}{n!} = e$.
Proof: Binomial, let $x_{n,k} = \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) < 1$
- Tut**: Suppose $a_n > 0, \sum a_n$ div. Let $s_n = \sum_{k=1}^n a_k$. Then $\sum \frac{a_n}{s_n}$ div.
Proof: Prove not Cauchy
- Tut**: Suppose $a_n > 0, \sum a_n$ conv. Then $\sum \frac{\sqrt{a_n}}{n}$ conv.
Proof: Let S_n be n -th partial sum. Consider S_n^2 , use Cauchy-Schwarz.
- Tut**: Suppose $a_n > 0$, for any $b_n \searrow 0, \sum a_n b_n$ conv. Then $\sum a_n$ conv.
Proof: Contradiction, consider $b_n = \frac{1}{\sqrt{a_n}}$.
- Tut**: If $a_n > 0, a_{m+n} \leq a_m + a_n \forall m, n \in \mathbb{N}$, then $\left\{\frac{a_n}{n}\right\}$ conv.
- Tut**: $\sum \sin n$ div.
Proof: Divergence test, construct a subsequence $\sin n_k \geq \frac{1}{2}$, so $\sin n_k \not\rightarrow 0$
- Tut**: Suppose $a_n > 0, \sum a_n$ conv, tail $r_n = \sum_{k=n}^{\infty} a_k$. Then $\sum \frac{a_n}{\sqrt{r_n}}$ conv.
Proof: $r_n \searrow 0$, seq of partial sums increasing and bounded above, MCT
- Suppose $a_n > 0, \sum a_n$ conv, tail $r_n = \sum_{k=n}^{\infty} a_k$. Then $\sum \frac{a_n}{r_n}$ div.
Proof: Show not Cauchy

C.8. Abel's summation formula

- Abel's summation formula**: Let $A_n = \sum_{k=1}^n a_k$. For any $q > p$,
$$\sum_{n=p}^q a_n b_n = A_q b_q - A_{p-1} b_p + \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}).$$

- Proof*: Write $A_n = A_n - A_{n-1}$, sub into summation.
Tip: Useful in proofs to show $\sum a_n b_n$ is Cauchy.
- Dirichlet's test**: Suppose (1) partial sums of $\sum a_n$ are bounded; (2) $b_n > 0, b_n \searrow 0$. Then $\sum a_n b_n$ is convergent.
- Abel's test**: Suppose (1) partial sums of $\sum a_n$ are convergent; (2) $\{b_n\}$ monotone and bounded. Then $\sum a_n b_n$ is convergent.

C.9. Rearrangement of series

- $\{b_n\}$ is a **rearrangement** of $\{a_n\}$ if $b_n = a_{f(n)}$, where $f: \mathbb{N} \rightarrow \mathbb{N}$ is bijection.
- Let $\{a_n\}$ be sequence s.t. $\sum a_n$ abs conv. If $\{b_n\}$ is a rearrangement of $\{a_n\}$, then $\sum b_n$ abs conv, and $\sum b_n = \sum a_n$.
- Riemann rearrangement thrm**: Let $\{a_n\}$ be sequence s.t. $\sum a_n$ conv conditionally. Then $\forall A \in \mathbb{R}, \exists$ rearrangement $\{b_n\}$ of $\{a_n\}$ s.t. $\sum b_n = A$.

D. Continuity

D.1. Limits of functions

- $x \in \mathbb{R}$ is **limit point** of $A \subseteq \mathbb{R}$ if $\forall \varepsilon > 0, A \cap (V_\varepsilon(x) \setminus \{x\}) \neq \emptyset$.
Equivalently, $\exists$ sequence $\{a_n\} \subseteq A, a_n \neq x$ s.t. $a_n \rightarrow x$.
- $\lim_{x \rightarrow c} f(x) = L$ if
$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad |f(x) - L| < \varepsilon$$
- Sequential criterion**: $\lim_{x \rightarrow c} f(x) = L$ if and only if for any sequence $\{a_n\} \subseteq A \setminus \{c\}, a_n \rightarrow c$, we have $\lim_{n \rightarrow \infty} f(a_n) = L$.
Tip: To compute limit of sequence, it suffices to compute limit of function.
- Algebraic operators**: Hold if limits exist
- Order preserving**: $f(x) \geq g(x)$ in a nbhd, then $\lim_{x \rightarrow c} f(x) \geq \lim_{x \rightarrow c} g(x)$.
- Squeeze**: Suppose $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = L$ and $f(x) \leq h(x) \leq g(x)$ in a nbhd. Then $\lim_{x \rightarrow c} h(x) = L$.
- Sign preserving**: If $\lim_{x \rightarrow c} f(x) = L > 0$, then $\exists C > 0$ s.t. $f(x) \geq C$ in a nbhd. If $\lim_{x \rightarrow c} f(x) = L < 0$, then $\exists C < 0$ s.t. $f(x) \leq C$ in a nbhd.
- $f: A \rightarrow \mathbb{R}$ **bounded** if $\exists C > 0$ s.t. $|f(x)| \leq C \forall x \in A$
If f not bounded, for every $n \in \mathbb{N}, \exists x_n \in [0, 1]$ s.t. $|f(x_n)| > n$. Then use Bolzano-Weierstrass on bounded $\{x_n\}$ to get conv subsequence.
- Tut**: Let $f: [0, 1] \rightarrow \mathbb{R}$ satisfy that for any $x \in [0, 1], \exists C > 0, \delta > 0$ s.t. $|f(y)| \leq C$ for all $y \in V_\delta(x) \cap [0, 1]$. Then f is bounded.

D.2. One-sided limits

- $\lim_{x \rightarrow c^-} f(x) = L$ if
$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c - \delta, c), \quad |f(x) - L| < \varepsilon$$
- $\lim_{x \rightarrow c^+} f(x) = L$ if
$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c, c + \delta), \quad |f(x) - L| < \varepsilon$$
- Boundedness, sequential criterion, order preserving, algebraic operators, squeeze theorem, sign preserving
- $\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$

D.3. Infinite limits

- $\lim_{x \rightarrow c} f(x) = +\infty$ if
$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad f(x) > L.$$
- $\lim_{x \rightarrow c} f(x) = -\infty$ if
$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad f(x) < -L.$$
- $\lim_{x \rightarrow c} f(x) = \infty$ if
$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad |f(x)| > L.$$

D.4. Limits at infinity

- $\lim_{x \rightarrow +\infty} f(x) = L$ if
$$\forall \varepsilon > 0, \quad \exists M > c, \quad \forall x > M, \quad |f(x) - L| < \varepsilon.$$
- $\lim_{x \rightarrow -\infty} f(x) = L$ if
$$\forall \varepsilon > 0, \quad \exists M < c, \quad \forall x < M, \quad |f(x) - L| < \varepsilon.$$
- $\lim_{x \rightarrow \infty} f(x) = L$ if $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = L$
- Sequential criterion, boundedness, order preserving, squeeze, sign preserving
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$

D.5. Continuous functions

- $f: A \rightarrow \mathbb{R}$ is **continuous** at $c \in A$ if
$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap V_\delta(c), \quad |f(x) - f(c)| < \varepsilon.$$

If c is limit point of A , then f cont at c if and only if $\lim_{x \rightarrow c} f(x) = f(c)$.
- Sequential criterion**: f cont at c if and only if for any sequence $\{a_n\} \subseteq A$ with $a_n \rightarrow c$,
$$\lim_{n \rightarrow \infty} f(a_n) = f(c).$$

Tip: Bring the limit in.
- Useful approx: $|\sin x| \leq |x|$ for $|x| \leq 1$
- Composition**: $f: A \rightarrow \mathbb{R}, g: B \rightarrow \mathbb{R}, f(A) \subseteq B$. Let c be limit pt of A and $\lim_{x \rightarrow c} f(x) = L$. Suppose $L \in B$ and g is continuous at L . Then
$$\lim_{x \rightarrow c} g(f(x)) = g(L).$$

Tip: Bring the limit in.
- Composition**: $f: A \rightarrow \mathbb{R}, g: B \rightarrow \mathbb{R}, f(A) \subseteq B$. Let c be limit pt of A and $\lim_{x \rightarrow c} f(x) = L$. Suppose further that $f(x) \neq L$ for $x \in A \cap (V_\delta(c) \setminus \{c\})$. If L is limit point of B and $\lim_{y \rightarrow L} g(y) = M$, then
$$\lim_{x \rightarrow c} g(f(x)) = M.$$

- $\lim_{x \rightarrow +\infty} x(\log(x+1) - \log x) = 1$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$$

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \frac{1}{2}$$

- Tip*: To deal with power function $f(x)g(x)$, rewrite as $e^{g(x) \log f(x)}$, then find limit of power.
- Tip*: By composition of functions, we can replace x with some $u \rightarrow 0$, then utilise above limits
- Boundedness**: If $f: [a, b] \rightarrow \mathbb{R}$ cont, then f is bounded.
- Extreme value**: If $f: [a, b] \rightarrow \mathbb{R}$ cont, then f attains absolute max & min.

D.6. Intermediate value theorem

- **Bolzano's IVT:** If $f: [a, b] \rightarrow \mathbb{R}$ is cont, $m = \min\{f(a), f(b)\}$, $M = \max\{f(a), f(b)\}$, then for any $L \in [m, M]$, there exists $c \in [a, b]$ s.t. $f(c) = L$.
Proof: Nested intervals whose intersection contains c
- **Location of roots:** If $f: [a, b] \rightarrow \mathbb{R}$ cont, $m = \min\{f(a), f(b)\}$, $M = \max\{f(a), f(b)\}$. If $m \leq 0$ and $M \geq 0$, then there exists $c \in [a, b]$ s.t. $f(c) = 0$.
Tip: Construct an appropriate function to apply IVT on (evaluate endpoints)
- If $f: [a, b] \rightarrow \mathbb{R}$ cont, $m = \min_{x \in [a, b]} f(x)$, $M = \max_{x \in [a, b]} f(x)$, then for any $L \in [m, M]$, there exists $c \in [a, b]$ s.t. $f(c) = L$. (Any value between max and min can be attained)
- Sufficient condition to show **interval**: If $S \subseteq \mathbb{R}$ satisfies that for any $x < y \in S$, $[x, y] \subseteq S$, then S is an interval.
- If $f: [a, b] \rightarrow \mathbb{R}$ cont, then the range $f([a, b])$ is a closed interval.
- **Preserve intervals:** If $f: I \rightarrow \mathbb{R}$ cont, then $f(I)$ is an interval.

D.7. Uniform continuity

- $f: A \rightarrow \mathbb{R}$ is **uniformly continuous** on A if
$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, y \in A, \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$
- *Trick:* Partition an interval into subintervals of length $< \delta$, apply above to each subinterval.
- **Preserve Cauchy:** If $f: A \rightarrow \mathbb{R}$ uniformly cont, $\{a_n\}$ Cauchy, then $\{f(a_n)\}$ is Cauchy.
- If $f: [a, b] \rightarrow \mathbb{R}$ cont, then f uniformly cont.
- **Continuous extension:** $f: (a, b) \rightarrow \mathbb{R}$ is uniformly continuous on (a, b) if and only if f can be extended to a continuous function on $[a, b]$, i.e., both $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$ exist.
- If limits at endpoint do not exist, then f is not uniformly continuous.
- **Tut:** If $f: \mathbb{R} \rightarrow \mathbb{R}$ uniformly cont, then there exist constants $A, B > 0$ s.t. $|f(x)| \leq A|x| + B \forall x \in \mathbb{R}$. (The function cannot grow faster than linearly.)
- $f: A \rightarrow \mathbb{R}$ is **Lipschitz** if there exists $K > 0$ such that $|f(x) - f(y)| \leq K|x - y| \forall x, y \in A$.
- Lipschitz $\implies$ uniform continuity
- Given $\alpha \in (0, 1)$, $f: A \rightarrow \mathbb{R}$ is α -**Hölder** if there exists $K > 0$ such that $|f(x) - f(y)| \leq K|x - y|^\alpha \forall x, y \in A$.
- Hölder $\implies$ uniform continuity
- **Tut:** Every α -Hölder function, with $\alpha > 1$, is constant.
- *Proof:* Partition $[x, y]$ into n subintervals, use given condition on each subinterval, then triangle inequality $|f(x) - f(y)| \leq \sum_{i=1}^n |f(x_{i+1}) - f(x_i)|$

D.8. Monotone functions

- If $f: I \rightarrow \mathbb{R}$ monotone, then one-sided limits exist at every interior point.
Proof: Consider $\sup_{x < c} f(x)$. Idea similar to proof of MCT.
- Suppose $f: I \rightarrow \mathbb{R}$ increasing, $c \in I$ interior point. TFAE:
 1. f is continuous at c ;
 2. $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$;
 3. $\sup_{x < c} f(x) = f(c) = \inf_{x > c} f(x)$.
- Suppose $f: I \rightarrow \mathbb{R}$ monotone, $c \in I$ interior point. The **jump** of f at c is
$$J_f(c) = \lim_{x \rightarrow c^+} f(x) - \lim_{x \rightarrow c^-} f(x).$$
- Suppose $f: I \rightarrow \mathbb{R}$ monotone, $c \in I$ interior point. Then f cont at c if and only if $J_f(c) = 0$.
- Suppose $f: I \rightarrow \mathbb{R}$ monotone. Then the set of discontinuous points of f on I is at most countable.
- **Continuous inverse:** Suppose $f: I \rightarrow \mathbb{R}$ strictly increasing, cont. Then f^{-1} is strictly increasing, cont on interval $I = f(I)$.

E. Basic Topology

E.1. Metric spaces

- (X, d) is a **metric space** if for all $x, y \in X$,
 1. $d(x, y) \geq 0$, where equality holds if and only if $x = y$; (non-negativity)
 2. $d(x, y) = d(y, x)$; (symmetry)
 3. for all $z \in X$, $d(x, y) \leq d(x, z) + d(z, y)$. (triangle inequality)
- $U \subseteq X$ is **open** if $\forall x \in U, \exists \delta > 0$ s.t. $V_\delta(x) \subseteq U$.
- $V \subseteq X$ is **closed** if V^c is open.
- Arbitrary union, finite intersection of open sets are open
- Arbitrary intersection, finite union of closed sets are closed
- A sequence $\{x_n\} \subseteq X$ **converges** to $x \in X$ if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad d(x_n, x) < \varepsilon.$$
- $x \in X$ is **limit point** of E if for any $\delta > 0$, $(V_\delta(x) \setminus \{x\}) \cap E \neq \emptyset$. Let E' denote the set of limit points of E .
- E is closed if and only if $E' \subseteq E$, i.e., E contains all its limit points.
- The **closure** of E is $\bar{E} = E \cup E'$.

- By above, $\bar{E}$ is closed, E is closed if and only if $E = \bar{E}$.
- $U \subseteq \mathbb{R}$ is open if and only if it is a disjoint union of finite or countably many open intervals.

E.2. General topology

- $(X, \mathcal{O})$ is a **topological space** if
 1. $\emptyset$ and X are open.
 2. For any subset $\mathcal{O}' \subseteq \mathcal{O}$, the union $\bigcup_{O \in \mathcal{O}'} O \in \mathcal{O}$.
 3. For any finite subset $\mathcal{O}' \subseteq \mathcal{O}$, the intersection $\bigcap_{O \in \mathcal{O}'} O \in \mathcal{O}$.
- $E \subseteq X$ is **closed** if $X \setminus E$ is open.
- **Induced topology** on $Y \subseteq X$: $U \subseteq Y$ is open relative to Y if $U = V \cap Y$ for some open $V \subseteq X$.

E.3. Compactness

- Let X be a topological space. $K \subseteq X$ is **compact** if any open cover of K , say $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, has a finite subcover, $K \subseteq \bigcup_{i=1}^n U_{\alpha_i}$.
- Let $K \subseteq Y \subseteq X$. Then K is compact in Y (with respect to the induced topology) if and only if K is compact in X .
- Let $K \subseteq X$ compact, $Y \subseteq X$ closed. Then $Y \cap K$ is compact.
- Let (X, d) be a metric space. Then compact subsets of X are closed.
- **Heine-Borel:** On $\mathbb{R}$, compact $\iff$ closed and bounded.
- **Nested cubes:** Let $\{K_i\}$ be a nested decreasing sequence of n -cubes. Then $\bigcap_{i=1}^\infty K_i \neq \emptyset$. If $\text{diam}(K_i) \rightarrow 0$, then the intersection is a singleton.
- **Bolzano-Weierstrass:** Any bounded sequence in $\mathbb{R}^n$ has convergent subsequence.
- **Heine-Borel:** On $\mathbb{R}^n$, compact $\iff$ closed and bounded.
- Let $\{K_\alpha\}_{\alpha \in \mathcal{A}}$ be compact subsets of metric space X such that for any finite subcollection, the intersection is non-empty. Then $\bigcap_{\alpha \in \mathcal{A}} K_\alpha \neq \emptyset$.
- Let $\{K_n\}$ be a nested decreasing sequence of non-empty compact subsets of metric space X . Then $\bigcap_{n=1}^\infty K_n \neq \emptyset$.

E.4. Sequential compactness

- $K \subseteq X$ is **sequentially compact** if every sequence in K has a convergent subsequence.
Equivalent to: any infinite subset $E \subseteq K$ has a limit point $\xi \in K$
- Compactness $\iff$ sequential compactness
Tip: Easier to use sequential compactness, since it involves sequences
- X is **separable** if it has a countable dense subset.
 $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ of open subsets of X is a **base** for X if for every $x \in X$ and for every open set $U \subseteq X$ such that $x \in U$, $\exists \alpha \in \mathcal{A}$ s.t. $x \in U_\alpha \subseteq U$.
- **Lemma 1:** Every separable metric space has a countable base.
Proof: $P = \{p_i\}$ countable dense, consider $\{V_r(p_i) : r \in \mathbb{Q}, i = 1, 2, \dots\}$.
- **Lemma 2:** If X is sequentially compact, then X is separable.
Proof: Fix $\delta > 0$. Having chosen $x_1, \dots, x_j$, choose x_{j+1} s.t. $d(x_i, x_{j+1}) \geq \delta$ for all $i = 1, \dots, j$. This process must terminate. Take $S = \bigcup_{n \in \mathbb{N}} S_{1/n}$, which is countable dense.
- Compact $\implies$ sequential compact
Proof: Contradiction, assume $E' \cap K = \emptyset$. Then $V_{\delta_x}(x) \cap E \subseteq \{x\}$ for every $x \in K$, so $K \subseteq \bigcup_{x \in K} V_{\delta_x}(x)$ is open cover.
- Sequential compact $\implies$ compact
Proof: For simple case, $\mathcal{A}$ is countable. Write $K \subseteq \bigcup_{i=1}^\infty U_i$. Contradiction, for every $n \in \mathbb{N}$, $K \not\subseteq \bigcup_{i=1}^n U_i$. Choose $x_n \in K$ s.t. $x_n \notin \bigcup_{i=1}^n U_i$. We get sequence $\{x_n\}$ in K , has conv subsequence $\{y_k = x_{n_k}\}$, let $y_k \rightarrow \xi \in K$. Since $\xi \in \bigcup_{i=1}^\infty U_i$, there exists $m \in \mathbb{N}$ such that $\xi \in U_m$. Since U_m open, $\exists \delta > 0$ s.t. $V_\delta(\xi) \subseteq U_m$. Since $y_k \rightarrow \xi$, choose k large enough such that $|y_k - \xi| < \delta$, so $y_k \in V_\delta(\xi) \subseteq U_m$. Note that $y_k = x_{n_k} \notin U_{m'}$ for all $m' \leq n_k$. In particular, $y_k \notin U_m$ (since $m \leq k$ implies $m \leq n_k$). This gives a contradiction. For general case, sequentially compact $\implies$ separable $\implies$ has countable base $\{U_n\}_{n \in \mathbb{N}}$. Write every U_α in terms of U_n .
- **Tut:** X metric space, $K \subseteq X$ compact, $F \subseteq X$ closed, $K \cap F = \emptyset$. Then there exists $\delta > 0$ s.t. $d(x, y) \geq \delta \forall x \in K, y \in F$.

E.5. Connectedness

- Let X be a topological space. X is **disconnected** if there exist disjoint non-empty open $A, B \subseteq X$ s.t. $X = A \cup B$. We call A, B a **separation** of X . If no separation of X exists, then X is **connected**.
- $S \subseteq X$ is **connected** if it is connected in the subspace topology. This means there do not exist disjoint non-empty sets $A, B \subseteq S$ such that $S = A \cup B$, and A and B are open in the subspace topology on S . Equivalently, there exist open sets U and V such that $A = S \cap U, B = S \cap V$. Explicitly, $S \subseteq X$ is disconnected if there exist open sets $U, V \subseteq X$ such that
 - $S \subseteq U \cup V$,
 - $S \cap U \neq \emptyset, S \cap V \neq \emptyset$,
 - $(S \cap U) \cap (S \cap V) = \emptyset$.
- *Tip:* To show connected, use contradiction, assume disconnected.
- On $\mathbb{R}$, connected $\iff$ interval.

E.6. Continuity

- Let X and Y be topological spaces. A map $f: X \rightarrow Y$ is **continuous** if for any $U \subseteq Y$ open, $f^{-1}(U)$ is open in X .
- **Preserve compactness:** If $K \subseteq X$ compact, then $f(K)$ compact.
- **Preserve connectedness:** If $C \subseteq X$ connected, then $f(C)$ connected.

E.7. Completeness

- A sequence $\{x_n\} \subseteq X$ is a **Cauchy sequence** if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad d(x_m, x_n) < \varepsilon.$$
- X is **complete** if every Cauchy sequence in X converges to an element of X .
- **Completion:** Let X^* denote the set of all Cauchy sequences in X :
$$X^* := \{P = \{p_n\} : \{p_n\} \subseteq X \text{ is a Cauchy sequence}\}.$$
 Define metric D on X^* : Let $P, Q \in X^*, P = \{p_n\}, Q = \{q_n\}$, then
$$D(P, Q) := \lim_{n \rightarrow \infty} d(p_n, q_n)$$
 and $P = Q$ if $D(P, Q) = 0$. (X^*, D) is a complete metric space.
- **Tut:** X is dense in X^*
- **Tut (fixed pt thrm):** (X, d) complete, $f: X \rightarrow X$ contraction. Then f has a unique fixed point.
Proof: Choose any $x_0 \in X$, define sequence $x_n = f(x_{n-1})$. Limit of $\{x_n\}$ should be fixed point.

F. Sequences and Series of Functions

F.1. Sequences of functions

- $f_n \rightarrow f$ **pointwise** if for any $x \in A$, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$
- $f_n \rightarrow f$ **uniformly** if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad \forall x \in A, \quad |f_n(x) - f(x)| < \varepsilon.$$
- **Sup norm** $\|f\| := \sup_{x \in A} |f(x)|$
- $f_n \rightarrow f$ uniformly if and only if $\lim_{n \rightarrow \infty} \|f_n - f\| = 0$.
- **Preserve continuity:** If $f_n \rightarrow f$ uniformly, each f_n cont, then f cont.
$$\lim_{n \rightarrow \infty} \lim_{x \rightarrow c} f_n(x) = \lim_{x \rightarrow c} \lim_{n \rightarrow \infty} f_n(x).$$
- $\{f_n\}$ is **Cauchy** if
$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad \|f_n - f_m\| < \varepsilon.$$
- $C[a, b]$ is complete.

F.2. Series of functions

- $\sum f_n(x)$ **conv uniformly** on E if sequence of partial sums $S_n(x) = \sum_{k=1}^n f_k(x)$ converges uniformly to a function $S(x)$ on E .
- If $\sum \|f_n\|$ converges, then $\sum f_n(x)$ converges uniformly.
Tip: Very useful to show uniform conv of series
Tip: To show $\sum \|f_n\|$ conv, bound $|f_n|$ by constant independent of x , then take sup to bound $\|f_n\|$, use comparison test.

F.3. Power series

- **Power series** $S(x) = \sum_{n=1}^\infty a_n x^n$
- Let $L = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$, define $R = 1/L$ as **radius of convergence**.
 1. If $L = +\infty$, then $\sum a_n x^n$ div for any $x \in \mathbb{R}$.
 2. If $L < +\infty$, then $\sum a_n x^n$ abs conv for $|x| < \frac{1}{L}$, div for $|x| > \frac{1}{L}$. Power series conv inside radius, div outside radius, not clear on boundary.
- Let $S(x) = \sum a_n x^n$ have radius of convergence $R > 0$. Then $S(x)$ cont on $(-R, R)$.
Proof: Show $\sum a_n x^n$ conv uniformly on $[-R', R'] \forall 0 < R' < R$.
- **Simple case:** Let $S(x) = \sum a_n x^n$ have radius of convergence R . If the series converges absolutely at $x = R$, then
$$\lim_{x \rightarrow R^-} S(x) = S(R).$$
 At boundary, if conv at endpoint, then limit function is cont.
- **Abel's thrm:** Let $S(x) = \sum a_n x^n$ have radius of convergence R . If the series converges (not necessarily absolutely) at $x = R$, then
$$\lim_{x \rightarrow R^-} S(x) = S(R).$$
- **Dini's thrm:** If $f_n: [a, b] \rightarrow \mathbb{R}$ cont, $f_1 \leq f_2 \leq \dots, f_n \rightarrow f$ pointwise, f cont, then $f_n \rightarrow f$ uniformly.
- **General:** Let X be a compact topological space. If $f_n: X \rightarrow \mathbb{R}$ cont, $f_1 \leq f_2 \leq \dots, f_n \rightarrow f$ pointwise, f cont, then $f_n \rightarrow f$ uniformly.
Proof: Let $g_n = f - f_n$, consider $E_n = \{x \in X : g_n(x) < \varepsilon\}$. Show $\{E_n\}$ is an open cover of X , $E_n \subseteq E_{n+1} \forall n \in \mathbb{N}$.
Tip: If f_n defined recursively, use MCT (increasing, bounded above) to show $\{f_n\}$ conv. Then take $n \rightarrow \infty$ to find f . Finally apply Dini's thrm.

Miscellaneous

Bag of tricks

- Split up equalities into inequalities: $x \leq y$ and $y \leq x$, $A \subseteq B$ and $B \subseteq A$
- Give yourself an epsilon of room:
To show $x = 0$, instead show $|x| \leq \varepsilon$ for all $\varepsilon > 0$
To show $x \leq y$, instead show $x \leq y + \varepsilon$ for all $\varepsilon > 0$
- Binomial theorem in inequalities
- $(a + b)^p \leq 2^p (a^p + b^p)$ for all $a, b, p \geq 0$
- Triangle inequality, insert a new term
- If working with ε , wonder to yourself if you can instead work with $\frac{1}{n}$, $n \in \mathbb{N}$. Sometimes, take $\varepsilon = 1$.
- If $x \in \mathbb{R}$, we can bound $n \leq x < n + 1$.
- First prove the result for a simple case, then decompose the general object into a combination of simple ones
- Dyadic partition, group terms of series
- Partition interval into subintervals (for uniform continuity)

Trigonometric formulae

Sum to product:

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Product to sum:

$$\sin \alpha \sin \beta = 1/2 [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = 1/2 [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = 1/2 [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = 1/2 [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$